$$\dot{x}_1 = -x_1 - \frac{x_2}{\ln\sqrt{x_1^2+x_2^2}}, \qquad \dot{x}_2 = -x_2 + \frac{x_1}{\ln\sqrt{x_1^2+x_2^2}}$$

has an equilibrium point at the origin.

(a) Linearize the system about the origin and show that the origin is a stable node of the linear system.

• 2.5 (a)

$$\frac{\partial f}{\partial x} = \begin{bmatrix} -1 + x_1x_2\alpha(x) & -\beta(x) + x_2^2\alpha(x) \\ \beta(x) - x_1^2\alpha(x) & -1 - x_1x_2\alpha(x) \end{bmatrix}$$

where

$$\alpha(x) = \frac{1}{(x_1^2+x_2^2)(\ln\sqrt{x_1^2+x_2^2})^2}, \quad \beta(x) = \frac{1}{\ln\sqrt{x_1^2+x_2^2}}$$

Noting that $\lim_{x\to 0} x_ix_j\alpha(x) = 0$ for $i,j = 1,2$ and $\lim_{x\to 0}\beta(x) = 0$, it can be seen that

$$\left.\frac{\partial f}{\partial x}\right|_{x=0} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \Rightarrow \text{the origin is a stable node}$$

(b) Transform the state equation into the polar coordinates

$$r = \sqrt{x_1^2+x_2^2}, \quad \theta = \tan^{-1}\left(\frac{x_2}{x_1}\right)$$

to obtain

$$\dot{r} = -r \Rightarrow r(t) = r_0e^{-t}$$